

C PROGRAMMING

BY USING SWITCH CASE...

main.c



Share

Run

```
1  #include <stdio.h>
2
3  int main() {
4      float num1, num2, result;
5      int choice;
6
7      printf("==== Basic Calculator ====\\n");
8      printf("1. Addition\\n");
9      printf("2. Subtraction\\n");
10     printf("3. Multiplication\\n");
11     printf("4. Division\\n");
12
13     printf("Enter your choice (1-4): ");
14     scanf("%d", &choice);
15
16     printf("Enter first number: ");
17     scanf("%f", &num1);
18
19     printf("Enter second number: ");
20     scanf("%f", &num2);
21
22     switch(choice) {
23         case 1:
```

```

9     printf("2. Subtraction\n");
10    printf("3. Multiplication\n");
11    printf("4. Division\n");
12
13    printf("Enter your choice (1-4): ");
14    scanf("%d", &choice);
15
16    printf("Enter first number: ");
17    scanf("%f", &num1);
18
19    printf("Enter second number: ");
20    scanf("%f", &num2);
21
22    switch(choice) {
23        case 1:
24            result = num1 + num2;
25            printf("Result = %.2f\n", result);
26            break;
27
28        case 2:
29            result = num1 - num2;
30            printf("Result = %.2f\n", result);
31            break;
32
33

```

main.c

Share

Run

```

28        case 2:
29            result = num1 - num2;
30            printf("Result = %.2f\n", result);
31            break;
32
33        case 3:
34            result = num1 * num2;
35            printf("Result = %.2f\n", result);
36            break;
37
38        case 4:
39            if(num2 != 0)
40                printf("Result = %.2f\n", num1 / num2);
41            else
42                printf("Error! Division by zero is not allowed.\n");
43            break;
44
45        default:
46            printf("Invalid choice! Please select between 1 and 4.\n");
47    }
48
49    return 0;
50 }

```

```

9  printf("2. Subtraction\n");
10 printf("3. Multiplication\n");
11 printf("4. Division\n");
12
13 printf("Enter your choice (1-4): ");
14 scanf("%d", &choice);
15
16 printf("Enter first number: ");
17 scanf("%f", &num1);
18
19 printf("Enter second number: ");
20 scanf("%f", &num2);
21
22 switch(choice) {
23     case 1:
24         result = num1 + num2;
25         printf("Result = %.2f\n", result);
26         break;
27
28     case 2:
29         result = num1 - num2;
30         printf("Result = %.2f\n", result);
31         break;
32 }

```

```

* ===== Basic Calculator =====
1. Addition
2. Subtraction
3. Multiplication
4. Division
Enter your choice (1-4): 1|
Enter first number: 12
Enter second number: 13
Result = 25.00

=== Code Execution Successful ===

```

The calculator program performs **Addition, Subtraction, Multiplication, and Division** on two numbers.

1. The user selects an operation from the menu.
2. The user enters two numbers.
3. The switch statement checks the selected choice.
4. The corresponding arithmetic operation is performed.
5. The result is displayed on the screen.
6. If an invalid choice is entered, an error message is shown.

Concepts Used:

- switch-case for operation selection
- scanf() for input
- printf() for output, Arithmetic operators (+, -, *, /)

